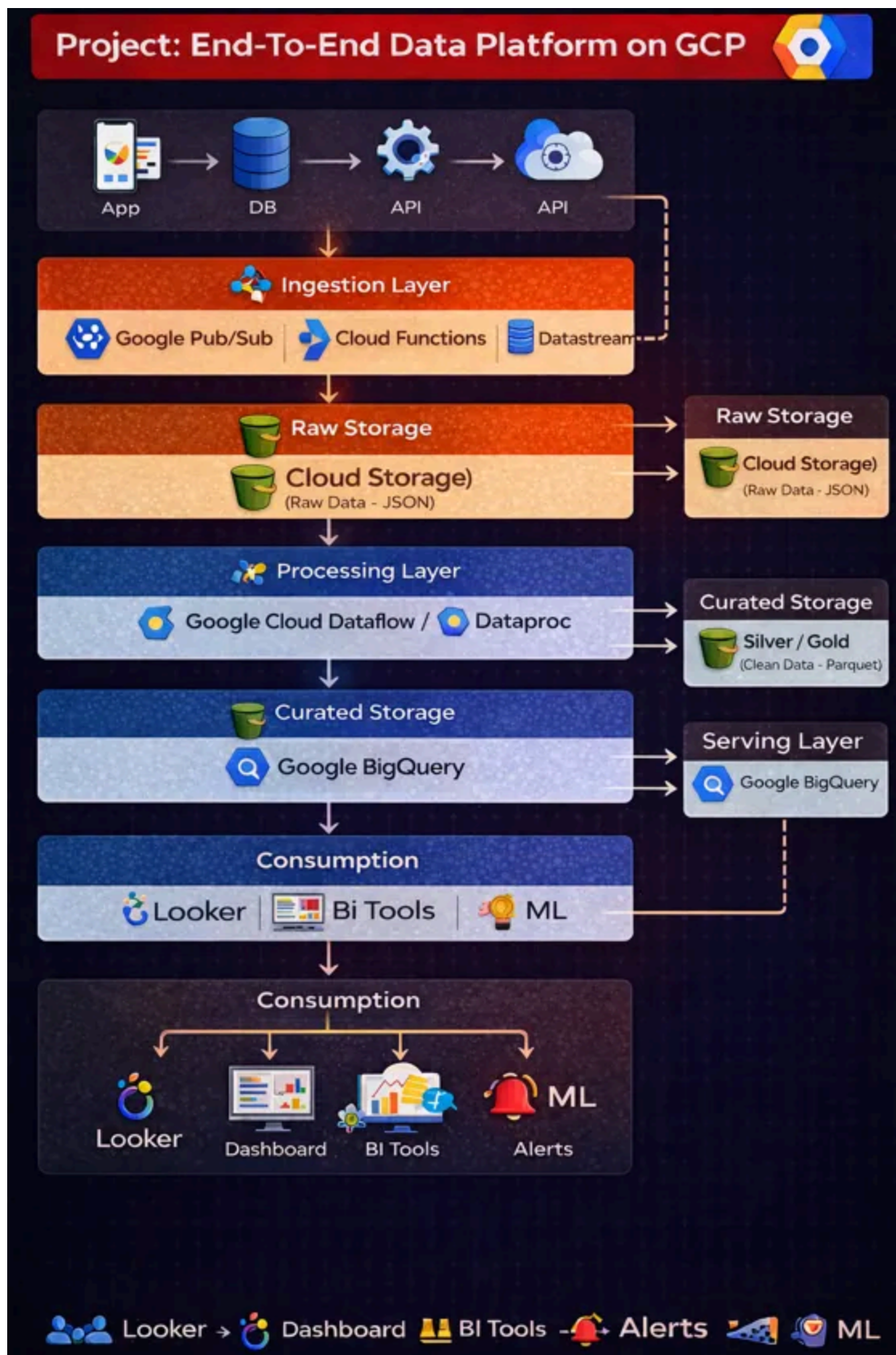




How Data Engineers Turn Business Requirements into Scalable Data Platforms



Anurag Sharma · 4 min read · May 4, 2026



In today's rapidly evolving AI-driven engineering landscape, the role of a Data Engineer is no longer limited to just writing ETL jobs or managing data infrastructure. Instead, it has expanded to encompass understanding business requirements, breaking them into actionable tasks, and accelerating implementation through AI-assisted development.

One of the biggest challenges engineers face is translating high-level Business Analyst (BA) requirements into structured, implementable data pipelines. Often, requirements come in abstract forms — “build a data platform,” “enable analytics,” or “support real-time insights.” The real skill lies in decomposing these into smaller, logical, and sequential tasks that can be developed, tested, and scaled independently.

In this blog, we take a practical, end-to-end approach.

We start with a single business goal:



Build a scalable data platform on GCP



Then, we break it down into 10 interconnected pipelines (tasks) — each representing a real-world step in a modern data engineering workflow:

- From data ingestion
- To validation and transformation
- To aggregation and warehouse loading
- All the way to monitoring and data access

Each pipeline is presented as:

- A clear BA requirement
- A technical implementation flow

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Why This Approach Matters

This structured breakdown helps you:

- Think like a Data Engineer + Architect
- Understand how real-world data platforms are built step-by-step

- Learn how to collaborate effectively with business stakeholders
- And leverage AI as a productivity multiplier, not just a coding tool

Business Context:

We are building a centralized analytics platform for a digital product company that collects:

- User activity logs
- Transactions
- External API data

The goal is to enable:

- Real-time monitoring
- Historical analytics
- ML readiness

Overall High-Level Flow

```
Sources (App, API, DB)
↓
Ingestion Layer (Pub/Sub, Cloud Functions, Datastream)
↓
Raw Storage (Cloud Storage - Bronze)
↓
Processing Layer (Dataflow / Dataproc)
↓
Curated Storage (Cloud Storage - Silver/Gold)
↓
Serving Layer (BigQuery)
```

↓

Consumption (Looker / BI Tools / ML / Alerts)

Pipeline 1: Data Ingestion (Batch + API + DB)

BA Requirement

We need to ingest:

- Application logs (batch upload)
- External API data (daily)
- Transaction DB changes (near real-time)

GCP Stack

- Cloud Storage (raw storage)
- Cloud Functions (API ingestion)
- Datastream (CDC from DB)

Flow

API → Cloud Functions → Cloud Storage
DB → Datastream → Cloud Storage
Logs → Cloud Storage upload

Sample Data Instruction

Generate mock API JSON:

```
{ "user_id": 101, "event": "login", "timestamp": "2026-01-01T10:00:00" }
```

Sample AI Prompt

“Generate a Google Cloud Function (Python) to fetch paginated API data and store it in Cloud Storage in JSON format with partitioning by date.”

Pipeline 2: Raw Data Validation

BA Requirement

Ensure all incoming data meets schema and quality standards.

GCP Stack

- Dataflow / Dataproc
- Cloud Storage
- Cloud Monitoring Alerts

Flow

```
Cloud Storage (raw) → Data Validation → Pass/Fail → Alerts
```

Sample Data

```
{ "user_id": null, "event": "", "timestamp": "invalid" }
```



Sample AI Prompt

“Write an Apache Beam (Dataflow) pipeline to validate schema, null checks, and timestamp formats with error logging.”

Pipeline 3: Data Lake (Bronze → Silver)

BA Requirement

Clean and standardize raw data into a structured format.

GCP Stack

- Dataflow / Dataproc
- Cloud Storage (partitioned parquet)

Flow

Raw (JSON) → Dataflow → Cleaned (Parquet)

Sample Transformation

- Convert timestamp → ISO
- Remove duplicates



Sample AI Prompt

“Generate PySpark or Apache Beam code to transform JSON data into partitioned Parquet format in Cloud Storage.”

Pipeline 4: Incremental Processing

BA Requirement

Process only new data (avoid full reloads).

GCP Stack

- Dataflow incremental processing
- Cloud Storage partitions

Flow

New Data → Incremental Job → Append to Silver

Sample Data

/data/year=2026/month=01/day=01/

Sample AI Prompt

“Write a Dataflow pipeline to process only newly added files in Cloud Storage using incremental logic.”

Pipeline 5: Data Enrichment

BA Requirement

Enhance data with external datasets (e.g., geo, user metadata).

GCP Stack

- Dataproc / Dataflow
- Cloud Storage

Flow

Silver Data + Lookup Data → Join → Enriched Data

Sample Data

```
{ "user_id": 101, "country": "US" }
```



Sample AI Prompt

“Create a PySpark job in Dataproc to join transactional data with a lookup dataset on user_id.”

Pipeline 6: Aggregation (Gold Layer)

BA Requirement

Create business metrics:

- Daily active users
- Transactions per hour

GCP Stack

- BigQuery

Flow

Enriched Data → Aggregation → Gold Tables

Example Output

```
date, active_users  
2026-01-01, 1200
```

Sample AI Prompt

“Write BigQuery SQL to calculate daily active users from event logs.”

Pipeline 7: Data Warehouse Load

BA Requirement

Load curated data into a warehouse for BI.

GCP Stack

- BigQuery
- Load from Cloud Storage

Flow

Cloud Storage Gold → BigQuery

Sample Tables

- users_metrics

- transaction_summary

Sample AI Prompt

“Generate BigQuery table schema and load command for partitioned Parquet data from Cloud Storage.”

Pipeline 8: Orchestration

BA Requirement

Automate the full pipeline with dependencies.

GCP Stack

- Cloud Composer (Airflow)

Flow

Ingestion → Validation → Processing → Aggregation → Load

Sample AI Prompt

“Create an Airflow DAG for orchestrating Dataflow jobs with retries and dependencies.”

Pipeline 9: Monitoring & Alerting

BA Requirement

Track failures, delays, and anomalies.

GCP Stack

- Cloud Monitoring
- Cloud Logging

Flow

Jobs → Logs → Alerts

Sample AI Prompt

“Configure GCP Cloud Monitoring alerts for Dataflow job failures and pipeline delays.”

Pipeline 10: Data Access Layer

BA Requirement

Provide easy access to data for analysts.

GCP Stack

- BigQuery
- Looker / Looker Studio

Flow

BigQuery → BI Dashboard

Sample AI Prompt

“Generate BigQuery external table DDL and sample queries for querying partitioned Cloud Storage data.”

Final Architecture Summary

```
[Sources]
API | DB | Logs
↓
Ingestion Layer
↓
Cloud Storage (Raw/Bronze)
↓
Validation Layer
↓
Processing (Dataflow / Dataproc)
↓
Cloud Storage (Silver → Gold)
↓
BigQuery
↓
BI / Alerts / ML
```

Why This is Strong for Portfolio

This setup demonstrates:

- Batch + streaming mindset
- Data lake architecture
- Orchestration
- Data quality
- Warehouse integration
- AI usage for building an E2E pipeline

Data Engineering

Data Engineer

Ai Data Engineering

Gcp Data Engineer

Big Data



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